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Testing Heavy Neutral Leptons with Multi-Messenger Observations of NGC 1068

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The active galactic nucleus NGC 1068, recently identified as a high-energy neutrino source by IceCube, exhibits a GeV γ-ray flux that exceeds predictions from opaque corona cascade models. This tension suggests the existence of non-attenuated energy transport mechanisms from the compact core. We investigate whether heavy neutral leptons (HNLs), coupled to the Standard Model through mixing with muon neutrinos, can provide such a channel. We calculate the production of HNLs in charged-pion and muon decays associated with hadronic interactions, explicitly tracking the mass-dependent polarization redistribution. We then derive the spectrum from visible $N \to \nu_\mu + e^+ + e^-$ decay channel and evaluate their secondary electromagnetic emission. We find that the predicted photon flux is at least 7 to 8 orders of magnitude below observational data once experimental bounds on $|U_{\mu N}|^2$ are applied. Even under an optimistic upper-limit estimate where particle transportation is neglected, this HNL channel is decisively ruled out as the origin of the γ-ray excess. Nevertheless, the polarization-resolved framework developed here remains applicable to broader studies.

Keywords: HNL; Inverse Seesaw; multi-messenger; NGC 1068

1. Introduction

Recent advancements in multi-messenger astronomy have opened new windows into astrophysical accelerators. Notably, the IceCube collaboration has identified the active galactic nucleus (AGN) NGC 1068 as a point source of neutrinos at energies around 1 TeV.[1] Furthermore, forthcoming instruments such as KM3NeT/ARCA, while still under construction, have observed a neutrino signal with energies possibly reaching 220 PeV.[2]

Within the Standard Model (SM), the non-zero but extremely small masses of active neutrinos point toward physics beyond the Standard Model (BSM). Consequently, Heavy Neutral Leptons (HNLs) have emerged as well-motivated candidates for both neutrino mass generation and dark matter.[3,4] Numerous terrestrial experiments are dedicated to searching for signals of BSM physics.[5] Although no definite

signals have been observed, they provide a stringent constraint on the parameter space of possible BSM theories, for HNLs with $3\,\mathrm{MeV} < m_N < 19.5\,\mathrm{MeV}$, the mixing angle with muon neutrino is tightly constrained to $|U_{\mu N}|^2 \lesssim 10^{-4}$.[6]

Moreover, the observed γ-ray flux of NGC 1068 exceeds theoretical predictions based on corona cascade models at energies around 10 GeV.[7,8] This suggests that the hidden core of NGC 1068 possesses mechanisms to transport additional electromagnetic energy outwards. Neutral, weakly interacting particles offer one possibility of transporting energy beyond an optically thick region before producing visible secondaries, provided such particles can be produced in hadronic decay chains. In this work we investigate whether heavy neutral leptons (HNLs) are viable candidates to mediate this process. A fraction of the particles may subsequently decay to charged leptons, which can radiate through conventional electromagnetic processes. Before NGC 1068, there are already attempts to combine multi-messenger observations to constrain BSM theories.[9–11] With NGC 1068, there are also studies focusing on constraining axion-like particles and investigating the possibility of neutrino decay.[12]

In this work, we test whether low-mass HNLs coupled to the muon sector can serve as this transport messenger. Unlike massless neutrinos, massive HNLs produce non-trivial polarization states in weak decays, and their subsequent decay distributions are governed by polarization. We formulate a polarization-resolved framework for both HNL production ($\pi^{\pm}$ and $\mu_{\pm}$ decays) and visible three-body decays ($N \to \nu_\mu + e^+ + e^-$), and apply the resulting secondary lepton spectra to the ambient environment of NGC 1068. This paper is structured as follows: In Section 2, we outline the basics of HNL theory originating from neutrino mass. In Section 3 and 4, we calculate its phenomenology properties, while taking polarization into account. In Section 5, we investigate the multi-messenger observation results of NGC 1068, try to explain the exceeding γ-ray flux. Conclusions are in Section 6.

2. ISS Model

To explain the tiny neutrino masses, several mechanisms known as seesaw mechanisms have been proposed. In the Type-I Seesaw model, the masses and mixing are given by

$$m_\nu = \frac{m_D^2}{M_R},\ m_N = M_R,\ |U|^2 = \left(\frac{m_D}{M_R}\right)^2 \tag{1}$$

where m_D is the Dirac mass of neutrino, and M_R is the Majorana mass of the added right-handed neutrino field. For a sufficiently natural m_D and M_R, the mixing angle $|U|^2$ is invariably too small, rendering direct observation practically impossible.

However, by introducing another fermionic field in the Inverse Seesaw (ISS),[13–15] the mass matrix becomes

$$\left(\overline{\nu_L}\ \overline{\nu_R^c}\ \overline{S^c}\right) \begin{pmatrix} 0 & m_D & 0 \\ m_D & 0 & M \\ 0 & M & \mu \end{pmatrix} \begin{pmatrix} \nu_L^c \\ \nu_R \\ S \end{pmatrix} \tag{2}$$

The parameter μ controls a small lepton-number-violating mass term. In the regime $\mu \ll M$ and $\mu \ll m_D$, the light neutrino mass and the heavy state mass follow the approximate expressions as

$$m_\nu = \mu \frac{m_D^2}{M^2},\ m_N = M,\ |U|^2 = \frac{m_\nu}{\mu} \tag{3}$$

The important feature is that the suppression of the active neutrino mass is now associated with the small parameter μ. Consequently the mixing can be larger than in a Type-I Seesaw model for comparable neutrino masses. The heavy states form a nearly degenerate pseudo-Dirac pair when μ is small, which in our calculating can be treated as a single Dirac fermion.

Because the calculation focuses on HNLs lighter than the charged pion decay threshold, their production and decay are governed by low-energy weak processes. The mixing parameter multiplies the weak-interaction amplitudes, while the nonzero mass modifies the available phase space and polarization content. These two effects can be separated conveniently in the following discussion.

3. HNL Production

Such HNLs interact only through their mixing with SM neutrinos, which implies that processes producing SM neutrinos will also produce HNLs, provided they are kinematically allowed.

Under AGN conditions, high-energy cosmic rays undergo $p-p$ and $p-\gamma$ interactions, which are thought to be the dominant sources of secondary particles.[16] Those processes create a large number of pions, whose decays subsequently produce high-energy cosmic rays, including neutrinos and γ-rays.[17]We consider the two channels most directly related to the muon-neutrino sector for HNL production: charged pion decay and muon decay.

3.1. *Pion decay*

The decay $\pi^+ \to \mu^+ + N$ is the HNL analogue of the process $\pi^+ \to \mu^+ + \nu_\mu$.[18]

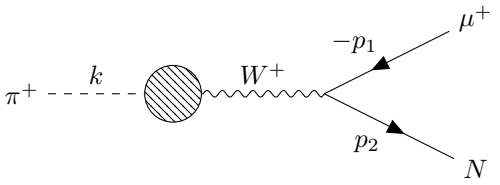


Fig. 1. Feynman diagram of π^+ decaying into μ^+ and massive HNLs.

By slightly modifying the process of SM pion decay, we find that the decay width

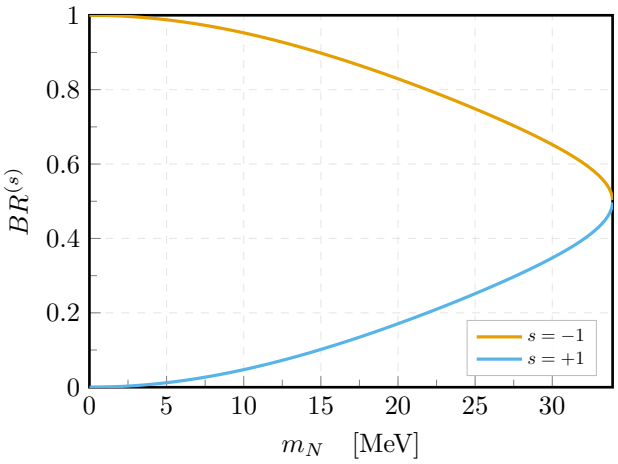


Fig. 2. Eq. (7) for different s. In pion decay, right-handed HNLs are more likely to be produced close to the phase space boundary.

to HNL is suppressed by the mixing angle and modified by a kinematic factor:

$$\frac{\Gamma_{\pi\to\mu N}}{\Gamma_{\pi\to\mu\nu}} = |U_\mu|^2 \times \rho_\pi(m_N) \tag{4}$$

where

$$\rho_\pi(m_N) = \frac{(m_\pi^2(m_\mu^2+m_N^2)-(m_\mu^2-m_N^2)^2)\sqrt{\lambda(m_\pi^2,m_\mu^2,m_N^2)}}{m_\mu^2(m_\pi^2-m_\mu^2)^2} \tag{5}$$

$$\lambda(x,y,z) = x^2+y^2+z^2-2xy-2yz-2zx \tag{6}$$

The function ρ_π encodes the competing effects of the HNL mass on the phase space.

It is useful to determine the polarization of the HNLs, since it will play a role in the angular dependence when working on the decay of HNL. In the pion rest frame we have

$$\Gamma_{(s)} \sim (m_\mu^2+m_N^2)m_\pi^2-(m_\mu^2-m_N^2)^2-s(m_\mu^2-m_N^2)\sqrt{\lambda(m_\pi^2,m_\mu^2,m_N^2)} \tag{7}$$

where s is the polarization of HNL, and $s=+1$ for right-handed polarized HNLs.

Since the parent pion is a spin-zero scalar, the decay is isotropic in its rest frame. However, the polarization of the massive HNL needs to be carefully tracked, as the subsequent Lorentz boost to the laboratory frame induces a mass-dependent polarization redistribution. For a pion spectrum $J(E_\pi)$, we get

$$Q_N^{(s)}(E_N) = \int_{\lambda^-}^{\lambda^+} J(\frac{E_N}{x})|U_\mu|^2\rho_\pi \times r_{(s)}(x,E_N) \times \frac{dx}{(\lambda^+-\lambda^-)\cdot x} \tag{8}$$

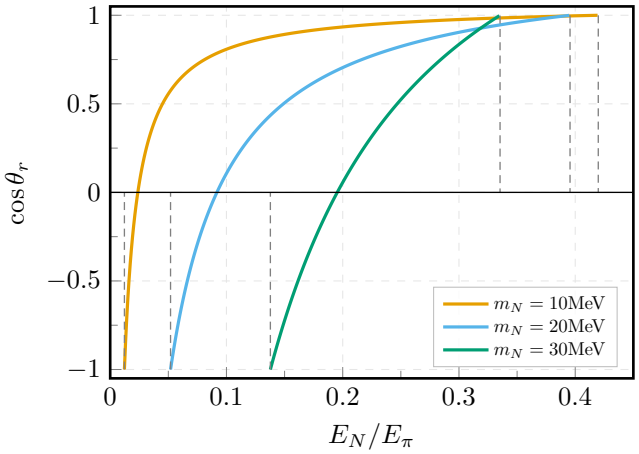


Fig. 3. Eq. (10) visualized with different m_N values. Eq. (10) determines how much helicity flip will happen during Lorentz boost. When m_N is small $\cos\theta_r \approx 1$, combined with Fig. (2), HNLs produced are almost left-handed; as m_N approaches the edge of phase space, the helicity flip is more significant and right-handed HNLs cannot be ignored.

where

$$\lambda^{\pm} = \frac{m_\pi^2 - m_\mu^2 + m_N^2 \pm \sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)}}{2m_\pi^2} \tag{9}$$

$$\cos\theta_r = \frac{x\,(m_\pi^2 - m_\mu^2 + m_N^2) - 2m_N^2}{x\sqrt{(1 - \frac{m_N^2}{E_N^2})\,\lambda(m_\pi^2, m_\mu^2, m_N^2)}} \tag{10}$$

$$r_{(s)}(x, E_N) = \frac{\Gamma_{(s)}\frac{1+\cos\theta_r}{2} + \Gamma_{(-s)}\frac{1-\cos\theta_r}{2}}{\Gamma_{(s)} + \Gamma_{(-s)}} \tag{11}$$

Here we assume $E_\pi \gg m_\pi$ to simplify the boost process. This resembles the SM neutrino case,[17] with modifications arising from HNL mass and helicity separation.

3.2. *Muon decay*

Muons can decay into HNLs via a process analogous to their standard decay into ν_μ. We find the decay width of $\mu^- \to Ne^-\bar{\nu}_e$ has the same form as the previously studied τ decay process,[19] since this process is symmetric between μ and τ, which is

$$\frac{\Gamma_{\mu\to N}}{\Gamma_{\mu\to\nu_\mu}} = |U_\mu|^2 \times \rho_\mu(m_N) \tag{12}$$

where

$$\rho_\mu(m_N) = -r_N^8 + 8r_N^6 - 24r_N^4\ln(r_N) - 8r_N^2 + 1, \quad r_N = m_N/m_\mu \tag{13}$$

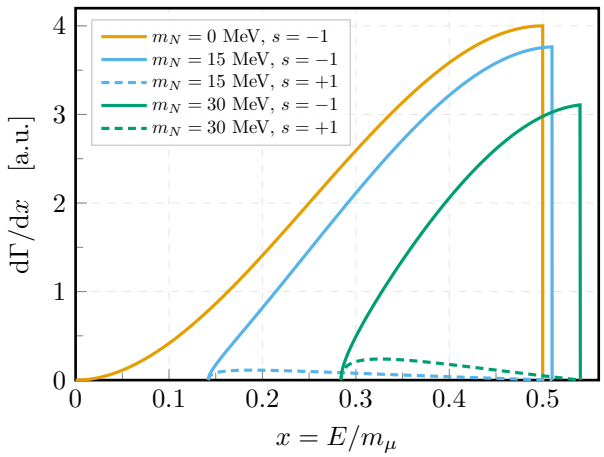


Fig. 4. Energy spectrum of HNL when muons decay into HNLs in muon rest frame. Normalized under $m_N = 0$, which basically resembles SM neutrino, with no right-handed product produced. When m_N increases, the shape of the spectrum remains roughly the same, but the maximum energy increases while the total decay width decreases, yielding a non-negligible fraction of right-handed HNLs.

The helicity-dependent spectrum of the HNL in the muon rest frame is given by

$$\frac{d\Gamma}{dE} = \frac{G_F^2}{24\pi^3}|U_\mu|^2 \times |\mathbf{p}| \cdot [3(m_\mu^2 + m_N^2)E - 4m_\mu E^2 - 2m_N^2 m_\mu - s \cdot |\mathbf{p}|(m_N^2 + 3m_\mu^2 - 4m_\mu E)] \tag{14}$$

Several examples are plotted in Fig. (4).

A fully spin-correlated treatment could retain the polarization of the parent muons and the angular relation between their spin and decay products. Here we neglect parent-muon polarization as a simplifying approximation. This approximation isolates the HNL mass and polarization dependence but should not be interpreted as a demonstrated consequence of complete magnetic depolarization in the source. For a given muon spectrum, we have

$$Q_N^{(s)}(E_N) = \int_{\frac{m_N^2}{m_\mu^2}}^1 \frac{dy}{2y}\, Q_\mu(\frac{E_N}{y})|U_\mu|^2\rho_\mu \cdot \int_{x^-}^{x^+} \frac{dx}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \cdot \left(\frac{1+\cos\theta_r}{2} g_N^{(s)}(x) + \frac{1-\cos\theta_r}{2} g_N^{(-s)}(x)\right) \tag{15}$$

where

$$x^+ = \frac{m_N^2 + m_\mu^2}{2m_\mu^2},\ x^- = \frac{1}{2}\left(y + \frac{1}{y}\cdot\frac{m_N^2}{m_\mu^2}\right) \tag{16}$$

$$\cos\theta_0 = \frac{y - x}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \tag{17}$$

$$\cos\theta_r = \frac{1}{y\sqrt{1 - \frac{m_N^2}{E_N^2}}}\left(\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}} + x\cos\theta_0\right) \tag{18}$$

where $g_N^{(s)}(x)$ is normalized as

$$\int_{\frac{m_N}{m_\mu}}^{x^+} dx \sum_{s=\pm 1} g_N^{(s)}(x) = 1 \tag{19}$$

4. HNL Decay

Kinematically, low-mass HNLs in the ISS model will exclusively decay into other leptons. Possible products are $\nu_\mu e^- e^+$ and $\nu_\mu \nu_l \bar{\nu}_l$. These processes have been well studied;[19–21] their decay widths are given by

$$\Gamma_{N\to 3\nu} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_\mu|^2 \tag{20}$$

$$\Gamma_{N\to \nu e\bar{e}} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_\mu|^2 \times C_1^e \tag{21}$$

where

$$C_1^e = \frac{1}{4}\left(1 - 4\sin^2\theta_W + 8\sin^4\theta_W\right) \tag{22}$$

To simplify the problem, we neglect the electron mass. The 3ν final state includes all neutrino flavor combinations.

An HNL produced in the central region can affect the electromagnetic signal only if its energy is ultimately transferred to visible charged particles, thus the branching ratio for an HNL to undergo a visible decay is

$$BR_{N\to \nu e\bar{e}} = \frac{C_1^e}{1 + C_1^e} \approx 0.11 \tag{23}$$

This results in substantial energy loss, as only approximately one-tenth of the HNL decays are observable.

The angular and energy distributions of electrons/positrons are

$$g_e^{(1)}(x,\cos\theta) = \frac{8}{C_1^e}\cdot x^2\cdot\{[(4-16s_W^2+64s_W^4)x-(1-4s_W^2+28s_W^4)]\cos\theta - (4-16s_W^2+64s_W^4)x+(3-12s_W^2+36s_W^4)\} \tag{24}$$

$$g_e^{(2)}(x,\cos\theta) = \frac{16}{C_1^e}\cdot x^2\cdot\{[(6-24s_W^2+32s_W^4)x-(3-12s_W^2+14s_W^4)]\cos\theta - (6-24s_W^2+32s_W^4)x+(3-12s_W^2+18s_W^4)\} \tag{25}$$

where $x = E_e/m_N$, $s_W = \sin\theta_W$. θ is the angle between the out-going electron and initial HNL spin. To get the total electron spectrum from a given HNL spectrum, we integrate over the HNL spectrum to obtain

$$Q_e^{(Lab)}(E) = BR_{N\to\nu e\bar{e}}\cdot\int_E^{+\infty} dE_N\, Q_N^{(s)}(E_N) \cdot\int_{\frac{E}{2E_N}}^{\frac{1}{2}}\frac{dx}{xE_N}\sum_{i=1,2} g_e^{(i)}(x, s(\frac{E}{xE_N}-1)) \tag{26}$$

Combining with 24 and 25, the remaining integral is

$$Q_e^{(Lab)}(E) = BR_{N\to\nu e\bar{e}} \cdot\int_0^1\frac{dy}{3y}\,Q_N^{(s)}(\frac{E}{y})[16y^3-27y^2+11-s\cdot(32y^3-63y^2+36y-5)] \tag{27}$$

where $y = E/E_N$, and θ_W is canceled while adding up Eq. (24) and Eq. (25). Under the mass-less limit of electron, chirality conservation ensures that the vector-axial-vector (V-A) interference terms are antisymmetric under charge conjugation.

5. HNL in NGC 1068

According to IceCube observations, NGC 1068 has been identified as a neutrino point source, and the dominant channel of neutrino production is charged pion decay.[8] By combining the analytical neutrino spectrum from pion decays[17] with theoretical models of NGC 1068,[8] we adopt the following parameterization as an approximation of the pion spectrum:

$$\Phi_{\pi^\pm}(E) = \Phi_0\cdot\left(\frac{E}{1\,\mathrm{GeV}}\right)^{-\Gamma}\cdot e^{-(E/E_c)^\beta} \tag{28}$$

where the best fit, which is visualized in Fig. (5), is

$$\Phi_0 = 4.7\times10^{-12}\,\mathrm{GeV}^{-1}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}, \Gamma = 0.1, E_c = 6.2\,\mathrm{GeV}, \beta = 0.3 \tag{29}$$

For HNLs to mediate energy transport from the opaque core to a transparent external environment, their decay length must be sufficiently large. We have

$$l_{dec} = \frac{E_N}{m_N}\,c\,\frac{\hbar}{\Gamma_{tot}} = \frac{192\pi^3 c\hbar}{G_F^2(1+C_1^e)}\cdot\frac{E_N}{m_N^6|U_\mu|^2} \tag{30}$$

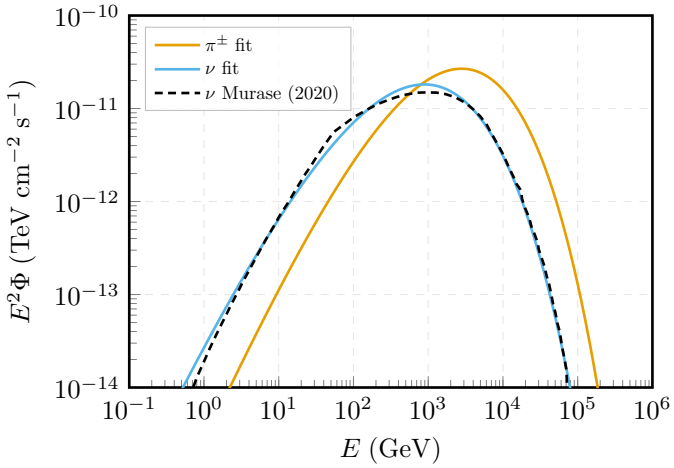


Fig. 5. Parametrization of the pion spectrum in NGC 1068 (Eq. (28)) for theoretical prediction.[8]

For demonstration, setting $|U_\mu|^2 = 10^{-5}$, $m_N = 20\,\mathrm{MeV}$, $E_N = 1\,\mathrm{TeV}$, yields $l_{dec} \approx 0.4\,\mathrm{pc}$, which implies that HNLs are unlikely to escape into interstellar space, while some decays will still be inside the optically opaque region. Thus, we assume all emitted energy escapes unattenuated as an optimistic approximation.

We refer to the parameters of the AGN torus model.[22] For our approximation we set the magnetic field strength to 139 mG, dust temperature to 1500 K, ambient density to $2.30 \times 10^5\,\mathrm{cm}^{-3}$, bolometric luminosity to $9.55 \times 10^{44}\,\mathrm{erg\,s}^{-1}$, and assume the electron-photon conversion happens at about 0.4 pc away from the AGN core. We use GAMERA to calculate the radiation generated by an input particle spectrum.[23] The result is shown in Fig. (7).

The principal outcome is the small magnitude of the predicted photon signal. Even with the mixing-dependent production normalization factored out for illustration, the spectrum lies roughly three to four orders of magnitude below the observationally relevant γ-ray scale in the comparison made here. Applying representative laboratory limits on the muon-neutrino mixing suppresses the production rate further. This supports the conclusion that the channel studied here cannot dominate the gamma-ray emission of NGC 1068 under the adopted assumptions.

The interpretation of the numerical suppression warrants two qualifications. First, the experimental upper bound on $|U_\mu|^2$ is mass dependent; the bound quoted for a particular low-mass interval should not automatically be extrapolated unchanged to the entire 10 MeV to 30 MeV range, though the difference in this range is minor. Second, an accurate observable flux from a finite external region must account for the probability of decay within that region, which depends on the HNL energy, mass, and mixing. If the radiation calculation effectively assumes that all relevant HNLs deposit their visible decay products in the chosen zone, it should be regarded as an optimistic estimate of the electromagnetic yield rather than a

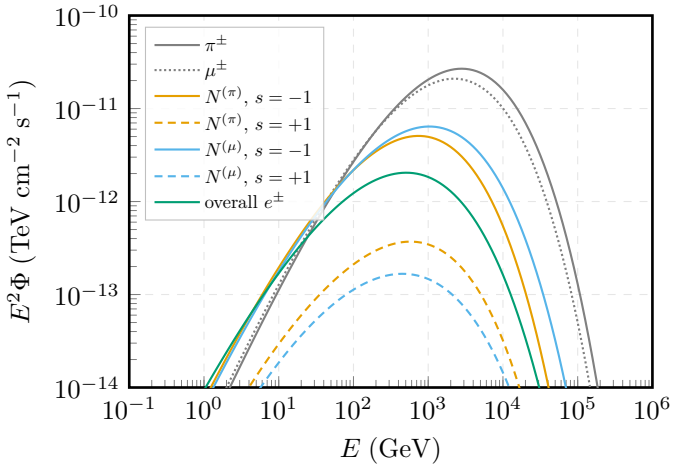


Fig. 6. HNL production source function from pion and muon decay for NGC 1068, using Eq. (28) as source pion spectrum. The polarization of muon is omitted. HNL mass is set to $m_N = 10\,\mathrm{MeV}$, and all HNL spectrums are scaled to $|U_\mu|^2 = 1$ to better show the physics. For $m_N = 10\,\mathrm{MeV}$, right-handed neutrino flux is 1 to 2 orders of magnitude smaller than left-handed one.

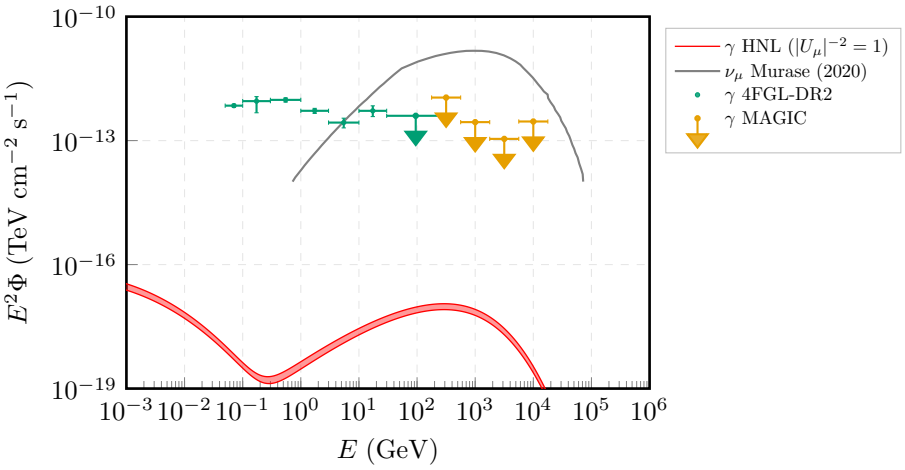


Fig. 7. Possible HNL generated photon spectrum compared with observations. $m_N = 10\,\mathrm{MeV}$ to $30\,\mathrm{MeV}$ is shown as a range. Even ignoring the mixing angle, which is smaller than 10^{-4}, the flux is 3 to 4 orders of magnitude smaller than the observational upper limit, which means it is at least 7 to 8 orders of magnitude smaller and therefore can be completely ignored in observation.

fully propagated flux prediction. These qualifications limit the scope of the numerical claim but do not motivate treating the present calculation as evidence for a detectable HNL signal.

6. Conclusions

In this article, we detailed the calculations of the production and decay processes of HNLs, assuming mixing with the muon neutrino sector. For $\pi^{\pm} \to N + \mu^{\pm}$ and $\mu^{\pm} \to N + e^{\pm} + \bar{\nu}_e$, we calculated the HNL spectrum separated by polarization, and applied Lorentz boost to the lab frame while taking polarization into account. Then we derived the electron/positron spectrum originating from HNL decay with different initial polarizations by adopting a representative external radiation environment for NGC 1068. Our modeling of NGC 1068 yields an HNL-associated γ-ray flux that is at least 7 to 8 orders of magnitude below current observational limits. Consequently, we disfavor the weakly coupled HNL channel as the primary mechanism for the anomalous γ-ray excess in this source. The limited branching ratio into visible channels and the specific secondary radiation processes, in addition to the small mixing angle, are the primary causes for the suppression of the HNL-induced γ-ray flux. The anomaly needs to be explained through a different process, most likely a refined model for the AGN environment on top of which is used in current electromagnetic cascade calculations.[8] Alternatively, energy transport mediated by other weakly interacting particles remains a possibility. Nevertheless, the polarization-resolved phenomenological framework developed herein remains a robust tool for evaluating similar BSM neutral-particle signatures in future studies.

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References

1. I. Collaboration*†, R. Abbasi, M. Ackermann, J. Adams, J. Aguilar, M. Ahlers, M. Ahrens, J. Alameddine, C. Alispach, A. Alves Jr *et al.*, *Science* **378**, 538 (2022).
2. *Nature* **638**, 376 (2025).
3. T. Asaka and M. Shaposhnikov, *Physics Letters B* **620**, 17 (2005), doi:https://doi.org/10.1016/j.physletb.2005.06.020.
4. A. Boyarsky, M. Drewes, T. Lasserre, S. Mertens and O. Ruchayskiy, *Progress in Particle and Nuclear Physics* **104**, 1–45 (January 2019), doi:10.1016/j.ppnp.2018.07.004.
5. S. N. Gninenko, *Phys. Rev. Lett.* **103**, 241802 (Dec 2009), doi:10.1103/PhysRevLett.103.241802.
6. D. Bryman and R. Shrock, *Physical Review D* **100**, 073011 (2019).
7. *The Astrophysical Journal* **883**, 135 (Sep 2019), doi:10.3847/1538-4357/ab3a51.
8. K. Murase, S. S. Kimura and P. Meszaros, *Physical review letters* **125**, 011101 (2020).
9. K. Hirata, T. Kajita, M. Koshiba, M. Nakahata, Y. Oyama, N. Sato, A. Suzuki, M. Takita, Y. Totsuka, T. Kifune, T. Suda, K. Takahashi, T. Tanimori, K. Miyano, M. Yamada, E. W. Beier, L. R. Feldscher, S. B. Kim, A. K. Mann, F. M. Newcomer, R. Van, W. Zhang and B. G. Cortez, *Phys. Rev. Lett.* **58**, 1490 (Apr 1987), doi:10.1103/PhysRevLett.58.1490.

10. T. Fischer, S. Chakraborty, M. Giannotti, A. Mirizzi, A. Payez and A. Ringwald, *Phys. Rev. D* **94**, 085012 (Oct 2016), doi:10.1103/PhysRevD.94.085012.
11. G. Magill, R. Plestid, M. Pospelov and Y.-D. Tsai, *Physical Review D* **98** (December 2018), doi:10.1103/physrevd.98.115015.
12. B. P. Pant, *Physical Review D* **109**, 063002 (2024).
13. R. N. Mohapatra, *Phys. Rev. Lett.* **56**, 561 (Feb 1986), doi:10.1103/PhysRevLett.56.561.
14. R. N. Mohapatra and J. W. F. Valle, *Phys. Rev. D* **34**, 1642 (Sep 1986), doi:10.1103/PhysRevD.34.1642.
15. A. Abada, V. De Romeri, J. Orloff and A. Teixeira, *The European Physical Journal C* **77**, 304 (2017).
16. K. Murase and F. W. Stecker, High-energy neutrinos from active galactic nuclei, in *The Encyclopedia of Cosmology*, (WORLD SCIENTIFIC, October 2023), p. 483–540.
17. S. R. Kelner, F. A. Aharonian and V. V. Bugayov, *Physical Review D—Particles, Fields, Gravitation, and Cosmology* **74**, 034018 (2006).
18. R. E. Shrock, *Phys. Rev. D* **24**, 1232 (Sep 1981), doi:10.1103/PhysRevD.24.1232.
19. K. Bondarenko, A. Boyarsky, D. Gorbunov and O. Ruchayskiy, *Journal of High Energy Physics* **2018** (November 2018), doi:10.1007/jhep11(2018)032.
20. A. Atre, T. Han, S. Pascoli and B. Zhang, *Journal of High Energy Physics* **2009**, 030–030 (May 2009), doi:10.1088/1126-6708/2009/05/030.
21. D. Gorbunov and M. Shaposhnikov, *Journal of High Energy Physics* **2007**, 015–015 (October 2007), doi:10.1088/1126-6708/2007/10/015.
22. E. Lopez-Rodriguez, C. Packham, T. J. Jones, R. Nikutta, L. McMaster, R. E. Mason, M. Elvis, D. Shenoy, A. Alonso-Herrero, E. Ramírez, O. González Martín, S. F. Hönig, N. A. Levenson, C. Ramos Almeida and E. Perlman, *Monthly Notices of the Royal Astronomical Society* **452**, 1902–1913 (2015), doi:10.1093/mnras/stv1410.
23. J. Hahn, GAMERA - a new modeling package for non-thermal spectral modeling, in *Proceedings of The 34th International Cosmic Ray Conference — PoS(ICRC2015)*, (2016), p. 917.